Notes on Computational Fluid Dynamics: General Principles

2.24 Summary of tensor algebra

Below is tensor algebra applied to Cartesian $(x,\ y,\ z)$ co-ordinates using: scalar s; vectors $\mathbf{a}$, $\mathbf{b}$; tensors $\mathbf{T}$, $\mathbf{S}$, $\mathbf{Y}$.

Products

- **Inner product of two vectors**, Sec. 2.3

$$s = \mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$$

- **Outer product of two vectors**, Sec. 2.8

$$\mathbf{T} = \mathbf{ab} = \begin{pmatrix} a_x b_x & a_x b_y & a_x b_z \\ a_y b_x & a_y b_y & a_y b_z \\ a_z b_x & a_z b_y & a_z b_z \end{pmatrix}$$

- **Inner product of vector and tensor**, Sec. 2.6

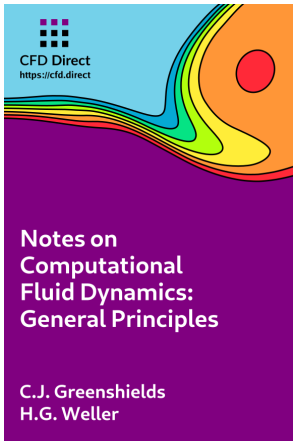
$$\mathbf{b} = \mathbf{a} \cdot \mathbf{T} = \begin{pmatrix} a_x T_{xx} + a_y T_{yx} + a_z T_{zx} \\ a_x T_{xy} + a_y T_{yy} + a_z T_{zy} \\ a_x T_{xz} + a_y T_{yz} + a_z T_{zz} \end{pmatrix}$$

- **Inner product of two tensors**, Sec. 2.8

$$\mathbf{Y} = \mathbf{T} \cdot \mathbf{S} \quad Y_{ij} = T_{ix} S_{xj} + T_{iy} S_{yj} + T_{iz} S_{zj} \quad (i, j = x, y, z)$$

- **Double inner product of two tensors**, Sec. 2.17

$$s = \mathbf{T} : \mathbf{S} = \begin{aligned} & T_{xx} S_{xx} + T_{xy} S_{xy} + T_{xz} S_{xz} + \\ & T_{yx} S_{yx} + T_{yy} S_{yy} + T_{yz} S_{yz} + \\ & T_{zx} S_{zx} + T_{zy} S_{zy} + T_{zz} S_{zz} \end{aligned}$$



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- **Cross product of two vectors**, first used in Sec. 3.3, produces a vector with components

$$\mathbf{a} \times \mathbf{b} = (a_y b_z - a_z b_y, a_z b_x - a_x b_z, a_x b_y - a_y b_x) \tag{2.70}$$

Tensors and operations

- **Transpose of a tensor**, Sec. 2.7

$$\mathbf{T} = \begin{pmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{pmatrix} \quad \mathbf{T}^{\mathrm{T}} = \begin{pmatrix} T_{xx} & T_{yx} & T_{zx} \\ T_{xy} & T_{yy} & T_{zy} \\ T_{xz} & T_{yz} & T_{zz} \end{pmatrix}$$

- **Symmetric and skew tensors**, Sec. 2.7

$$\mathbf{T} = \underbrace{\frac{1}{2}(\mathbf{T} + \mathbf{T}^{\mathrm{T}})}_{symmetric} + \underbrace{\frac{1}{2}(\mathbf{T} - \mathbf{T}^{\mathrm{T}})}_{skew} = \operatorname{sym}\mathbf{T} + \operatorname{skw}\mathbf{T}$$

- **Trace of a tensor**, Sec. 2.10

$$\operatorname{tr}(\mathbf{T}) = T_{xx} + T_{yy} + T_{zz}$$

- **Identity tensor**, Sec. 2.8

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{I} \cdot \mathbf{T} \equiv \mathbf{T} \cdot \mathbf{I} \equiv \mathbf{T}$$

$$\mathbf{T} : s\mathbf{I} \equiv s \operatorname{tr}(\mathbf{T})$$

- **Deviatoric and spherical tensors**, Sec. 2.10

$$\mathbf{T} = \underbrace{\mathbf{T} - \frac{1}{3}(\operatorname{tr}\mathbf{T})\,\mathbf{I}}_{deviatoric} + \underbrace{\frac{1}{3}(\operatorname{tr}\mathbf{T})\,\mathbf{I}}_{spherical} = \operatorname{dev}\mathbf{T} + \operatorname{sph}\mathbf{T}$$

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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